



Gamification in collaborative learning: synthesizing evidence through meta-analysis

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Abstract

Gamification has emerged as a potent tool for enhancing collaborative learning experiences, both in face to face and online environments. Based on real-world evidence of its effectiveness in various fields like online shopping and healthy habits, the efficacy of gamification in improving engagement has also sparked a great interest among scholars in the education field. This meta-analysis aims to scrutinize the efficacy of gamification in collaborative learning settings across educational levels, encompassing both K-12 and higher education environments. A comprehensive search across prominent academic databases (i.e., WOS SSCI, Scopus, and Google Scholar) yielded 35 relevant studies, comparing outcomes between gamified and non-gamified learning conditions. Among the 35 studies, 28 studies examined cognitive outcomes of learning (3296 participants), and 22 studies examined attitudinal outcomes of learning (2862 participants). For cognitive outcomes, the overall effect size was $d=0.875$ ($SE=0.183$, $p<0.0001$, 95% CI [0.517, 1.232]), indicating a large effect. Similarly, significant effects were observed in attitudinal learning outcomes or socio-emotional domains, with an effect size of $d=0.748$ ($SE=0.159$, $p<0.0001$, 95% CI [0.436, 1.060]), signifying a moderate to large impact. Our findings underscore the substantial positive influence of gamification on cognitive and attitudinal learning outcomes when integrated into collaborative learning contexts. Furthermore, we provide insights into the demographic characteristics of the included studies, offering a nuanced understanding of participant profiles and collaborative learning contexts. Additionally, we discuss avenues for future research and acknowledge the limitations inherent in this meta-analysis. In conclusion, this meta-analysis affirms the efficacy of gamification in enhancing both cognitive and attitudinal dimensions of collaborative learning, thereby advocating for its strategic integration into educational practices.

Keywords Meta-analysis · Collaborative learning · Teamwork · Gamification

Introduction

Collaborative learning (CL) is an educational method where students collaborate in small groups to tackle problems, accomplish tasks, or produce something new (Meske et al., 2017; Slavin, 1980). Within this context, learners are not only intellectually challenged from both social and emotional perspectives as they listen to various viewpoints and are compelled to articulate and justify their own ideas (Su & Guo, 2024). In a collaborative learning environment, students interact with peers, present and defend their ideas, exchange diverse perspectives, question different conceptual frameworks, and actively participate in the learning process rather than relying solely on an expert's or textbook's viewpoint (Laal & Laal, 2012).

Numerous benefits of CL have been detected concerning the educational field (D. W. Johnson & Johnson, 1999; Panitz, 1999), including benefits from social, psychological, academic, and educational assessment perspectives (Laal & Laal, 2012). On the other hand, collaborative learning also faces several challenges, drawbacks, and limitations that can impede its effectiveness when being put into practice (Chiriac & Granström, 2012; De Hei et al., 2015; Freeman & Greenacre, 2011; Janssen et al., 2007; Kreijns et al., 2003; Le et al., 2018a; Li & Campbell, 2008; Panitz, 1999; Pauli et al., 2008). Among the various strategies employed to enhance students' learning experiences in collaborative environments, gamification has emerged as a particularly influential educational phenomenon (da Rocha Seixas et al., 2016).

Gamification usually involves integrating game mechanics, or game design elements, such as points, badges, levels, challenges, and rewards, into activities or processes that are non-game context (Zainuddin et al., 2020a, 2020b). Given this, rapid number of research has been conducted on implementing gamification to enhance collaborative learning experience or help eliminate the challenges of effectively putting collaborative learning into practice. However, despite the growing enthusiasm among researchers for integrating gamification into collaborative learning activities and designs, there remains a lack of comprehensive meta-analyses examining its actual outcomes and effects in these environments. Therefore, this study aims to fill this gap by investigating the extent to which gamification is effective in helping learners achieve various learning outcomes in collaborative settings. Additionally, this meta-analysis seeks to understand the trends in scholarly research on gamification in collaborative learning, providing empirical evidence about the explored and unexplored areas concerning learning outcomes. This study also delves into rationale why certain aspects of learning outcomes or learning, in general, are rarely investigated and makes some recommendations for further studies. This study specifically answered the following questions:

1. What types of learning outcomes did researchers investigate when using gamification in collaborative learning?
2. To what extent did gamification influence students' learning outcomes within collaborative learning settings?
3. How did the length of experiment and education level influence the outcomes of the studies?

What is collaborative learning

The fundamental principle of collaborative learning is based on Vygotsky's works on sociocultural theory of cognitive development that asserted that knowledge "... is not a give attribute but a social artifact constructed in the process of social interaction among communities of knowledgeable peers" (Yang, 2023, p.722). A comprehensive definition of collaborative learning has been developed to aid research and practice. Philosophically, traditional collaborative learning involves creating an environment where students can effectively bridge the gaps between their own knowledge and that of their instructor (Bruffee, 1999). This early definition emphasizes the idea that knowledge is co-constructed, with all members of the learning community, whether instructors or students, playing equal roles in building the body of knowledge. Collaborative efforts can vary in group size, composition, nature of collaboration, and communication methods (Dillenbourg, 1999). Another influential definition describes collaborative learning as processes that foster interaction, cooperation, and learning in small groups (two to five learners), ensuring that each member contributes to and benefits from the group's learning (Johnson & Johnson, 1999).

To give a slight description, Laal and Laal (2012) describe collaborative learning as an educational approach where groups of learners work together to solve problems, complete tasks, or create products. This process challenges learners emotionally, academically, and socially, encouraging them to articulate and defend their ideas while remaining open to others' perspectives (S. C. Tan et al., 2022). Similarly, Jeong and Hmelo-Silver (2016) define collaborative learning as involving two or more people working together towards shared goals and solving problems, often requiring joint and balanced participation from all members. In the process, learners are more likely to have agency, accountability, independence, and responsibility towards their learning progress (Tan et al., 2022). These definitions frame collaborative learning as an umbrella term encompassing various instructional strategies for small group learning, such as cooperative learning, team-based learning, peer tutoring, study groups, project-based learning, problem-based learning, and learning communities (Yang, 2023). Based on the previous concept and definition, for our study, we define collaborative learning as learning processes where learners are encouraged to work together in a group toward shared goals. Our journal selection criteria will follow this definition, focusing on studies where collaborative learning activities involve learners working together in groups to achieve shared goals in specific learning contexts.

Collaborative learning in practice

From the implementation standpoint, collaborative learning activities are always led by open-ended tasks that determine the shared goals that groups attempt to achieve, and the process are likely to be loosely structured (Yang, 2023). These open-ended tasks encourage creativity and critical thinking, allowing students to explore various solutions and approaches. In the same paper, common strategies for collaborative

learning include peer tutoring, where students teach each other; collaborative writing, where group members contribute to a single document; project-based learning, which involves students working together on extensive projects; group quizzes, which promote collective problem-solving; and group discussions, fostering a rich exchange of ideas. The size of learner groups can vary significantly depending on the specific goals and requirements of the activities. In addition to these traditional methods, collaborative learning sometimes goes beyond small, fixed groups to include interactions between multiple groups, enhancing the learning experience through a broader range of perspectives and contributions (Jeong & Hmelo-Silver, 2016). This inter-group collaboration can lead to a more comprehensive understanding of the subject as students benefit from diverse viewpoints and expertise. Technological advancements have further expanded the possibilities for collaborative learning. Some studies explore the concept of Mass Collaborative Learning (MCL), where collaboration takes place as a large number of dispersed, self-directed individuals share their knowledge, information, data, and experiences, typically through ICT platforms (Chen et al., 2021; Zamiri & Camarinha-Matos, 2019). This process involves collective action where knowledge is continuously created, shared, and developed collaboratively. Mass-scale collaboration has become increasingly popular, exemplified by platforms like Wikipedia, where users collectively build and edit content, and Scratch, where users can develop and modify games (Tan et al., 2022). These platforms demonstrate how technology can facilitate extensive collaborative efforts, allowing learners from around the world to contribute to and enhanced projects. Moreover, technology has opened new avenues for collaboration between humans and artificial intelligence. For example, learners can now work alongside robots or interact with chatbots, which can provide instant feedback, suggest resources, and offer different perspectives, thereby enriching the collaborative learning experience (Miyake et al., 2012).

Benefits and challenges of collaborative learning in improving learning outcomes

Scientific research has documented numerous benefits associated with the implementation of collaborative learning, which can be categorized into four main areas: social benefits, psychological benefits, academic benefits, and alternative assessment methods for both students and teachers (D. W. Johnson et al., 2007; Laal & Ghodsi, 2012; Le et al., 2018a; Slavin, 1996). Socially, collaborative learning fosters the development of essential interpersonal skills such as communication, teamwork, and conflict resolution. Psychologically, it enhances students' self-esteem, motivation, and overall mental well-being by creating a supportive learning environment. Academically, it promotes deeper understanding, critical thinking, and retention of information through active engagement and peer interaction. In addition, collaborative learning offers alternative assessment techniques, allowing for a more comprehensive evaluation of student learning and progress beyond traditional tests and exams, such as group assessment and group assessment combined with intragroup peer assessment (Boud & Bearman, 2024; Meijer et al., 2020) or classroom assessments (Andrade et al., 2020a, 2020b). This approach enables students to develop

crucial cognitive skills, including analysis, synthesis, and problem-solving, alongside pro-social behaviors like empathy, cooperation, and helping others. These skill sets are particularly beneficial for higher education students, who are being prepared for the job market, where collaborative and team-based environments are becoming increasingly common and essential (De Hei et al., 2015).

However, collaborative learning also faces significant challenges and limitations that can hinder its effectiveness. Kreijns et al. (2003) highlight a key issue: the assumption that simply creating a space for interaction will automatically foster social engagement and learning. Several intrinsic factors contribute to these challenges, including learners' inability to collaborate effectively, free-riding behaviors, relationships with peers, and varying levels of competence (Le et al., 2018b). These internal dynamics can further exacerbate problems, as learners struggle to navigate and address issues related to teamwork and performance (Hadwin et al., 2018). Lecturers often overlook the importance of fostering group cohesion, trust, respect, and a sense of belonging among students, which are crucial for effective collaboration. Additionally, teachers frequently struggle with forming effective groups, due to limited knowledge of research and theoretical perspectives on collaborative learning, and difficulty translating theoretical insights into practical applications. This lack of clarity and knowledge affects the implementation of collaborative learning, resulting in suboptimal supervision, especially when teachers are overwhelmed by having to manage too many groups simultaneously (De Hei et al., 2015).

Further complicating the adoption of collaborative learning are various fears and uncertainties among educators. Research highlights that teachers often fear losing control over the classroom and feel insecure due to past unsuccessful experiences with collaborative learning (Panitz, 1999). Concerns about covering the entire curriculum, unfamiliarity with assessment techniques, and student resistance to collaborative learning due to their lack of experience with such methods also pose significant barriers (De Hei et al., 2015). From the students' perspective, challenges include unequal participation and a lack of collaborative and communicative skills, which can lead to low-quality interactions and hinder both group and individual learning (Freeman & Greenacre, 2011; Janssen et al., 2007; Li & Campbell, 2008; Pauli et al., 2008). Teachers, on the other hand, face difficulties in designing suitable group tasks, managing class time, and ensuring productive collaboration (Gillies & Boyle, 2008; van Leeuwen & Janssen, 2019). Research by Barron (2003) on primary school children's collaborative interactions revealed issues such as poor coordination and lack of attentiveness to peers' opinions, which further undermine the effectiveness of collaborative learning. Thus, both organizational and assessment challenges significantly impact the successful implementation of collaborative learning in educational settings (Chiriac & Granström, 2012; Le et al., 2018a).

Gamification in enhancing learning

Gamification has been considered a practical approach to engaging students in learning. The concept of gamification, which involves applying game elements and design techniques in non-game contexts, was first defined in the early 2000s (Braga, 2022;

Deterding et al., 2011). Today, gamification is widely used across various fields such as business, marketing, health, technology design, and education, and is seen as a crucial strategy for promoting student participation and engagement (L. Johnson et al., 2016). Research into gamification in education has yielded promising insights. A comprehensive examination by Huang et al. (2020) highlighted gamification's effectiveness across various educational domains, encompassing behavioral, affective, and cognitive dimensions. In a meta-analysis conducted by Bai et al. (2020), they studied a total of 30 independent interventions involving 3202 participants derived from 24 quantitative studies investigating the impact of gamification on student academic performance in diverse educational environments. They realized that gamification can be used to promote learners' goal settings, fulfil their needs for recognition, offer feedback on both the individual learner's performance and the performance of their peers, and provide novelty effects to learners if gamification interventions are in short terms. Besides conventional education contexts, gamification has also played an active role in enhancing learning in online environments. Multiple studies have been conducted to show how gamification can be implemented to facilitate online learning and teaching. In higher education field, gamification has been applied to facilitate or enhance online learners' effectiveness in nursing education (Inangil et al., 2022), computer science education (Khaleel et al., 2019), modern management and information technology education (Pakinee & Puritat, 2021), physics education (Ahmed & Asiksoy, 2021), medical education (Do et al., 2023), adult education (Ng & Lo, 2022), and it has also been used to improve curriculum design (Chung & Lin, 2022) and MOOCs (Metwally & Yining, 2017; Romero-Rodriguez et al., 2019).

Naturally, the widespread adoption of gamification in educational settings also encompasses the realm of collaborative learning. A meta-analysis conducted by Ho et al. (2022) showcased a noteworthy enhancement in cognitive learning outcomes when gamification was integrated into collaborative learning activities. This underscores the potential of gamified approaches to positively influence educational outcomes. Further to contextualize these findings, studies such as that of Lou et al. (2001) underscored the pivotal role of peer collaboration in bolstering group performance, highlighting its superiority over individual learning achievements. These collective findings underscore the multifaceted benefits of integrating gamification and collaborative learning strategies, advocating for their broader adoption in educational contexts. Given this, it is essential for scholars to have a systematic review of the emerging research finding regarding gamification implementation into collaborative learning and evaluate the consequent outcomes and its overall effectiveness.

Methods

Database searched and search string

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure rigor and transparency in our study selection process (Page et al., 2021). Comprehensive searches were conducted across major

scholarly databases, including Web of Science SSCI, Scopus, and Google Scholar, covering the publication period from 2011 to 2024. WOS and Scopus were selected because they are well known as multidisciplinary databases that include high-quality peer-reviewed journals. By constructing the corpus from high quality peer-reviewed publications, we hoped to ensure the credibility of the articles selected. In addition, Google Scholar was included to capture articles that may not be indexed in WOS or Scopus but are still highly relevant to our research topic. We initially conducted data collection using WOS and Scopus due to their established reputation for high-quality, peer-reviewed content. Following this, we expanded our search to include Google Scholar to ensure a more comprehensive review. Google Scholar offered access to a broader range of publications, including those from diverse fields and emerging areas of study, which helped us to include a wider array of relevant studies in our analysis.

This time frame was chosen to encompass the evolution of gamification research, from its seminal definition coined by Deterding et al. (2011) to the latest developments in the field up to 2024 (Deterding et al., 2011). In WOS and Scopus, we utilized the citation report feature to retrieve and compile the list of references. This functionality facilitated the organization of results into an Excel file. Meanwhile, for Google Scholar, we utilized *Harzing's Publish or Perish* software, executing an identical search query to retrieve and export the findings into a more manageable Excel format. For ensuring the replicability of study, we elaborated on the search process of our selected references (see Fig. 1). Our search query (with Boolean expression) involved *gamif** AND *collaborat** OR *cooperat** OR *collective** OR *team* AND *learning*. In total, the search query found 1648 papers within the selected time of publication (2011–2024).

Inclusion and exclusion criteria

Our research questions shaped the inclusion and exclusion criteria of the corpus. To achieve our ultimate goal, we specifically selected studies that meet the following criteria: (1) published between January 1 2011 and October 31 2024, (2) published in a peer-reviewed journal indexed by Google Scholar, Scopus, and WOS (Web of Science), (3) written in English, (4) primarily focus on gamification studies, (5) must be an empirical study (review or commentary, systematic literature review, or meta-analysis were excluded), (6) full-paper were available, (7) must be completed in the context of K-12 or higher education, (8) must be set within the context of collaborative learning or teamwork activities, and (9) must be empirical studies involving quasi-experimental with robust method (comparison between treatment and control groups) and sufficient statistical information. Before applying the inclusion and exclusion criteria, we checked the relevance of the retrieved papers. In this stage, we excluded the papers that were non-English written, duplicated, and had irrelevant topics ($n=551$). Following that, the screening process was performed. In the screening phase, we employed the inclusion and exclusion criteria that were established based on our research questions. We removed 954 papers due to our exclusion criteria: (1) non-peer reviewed journal, (2) non-empirical study, (3) non-full paper, and

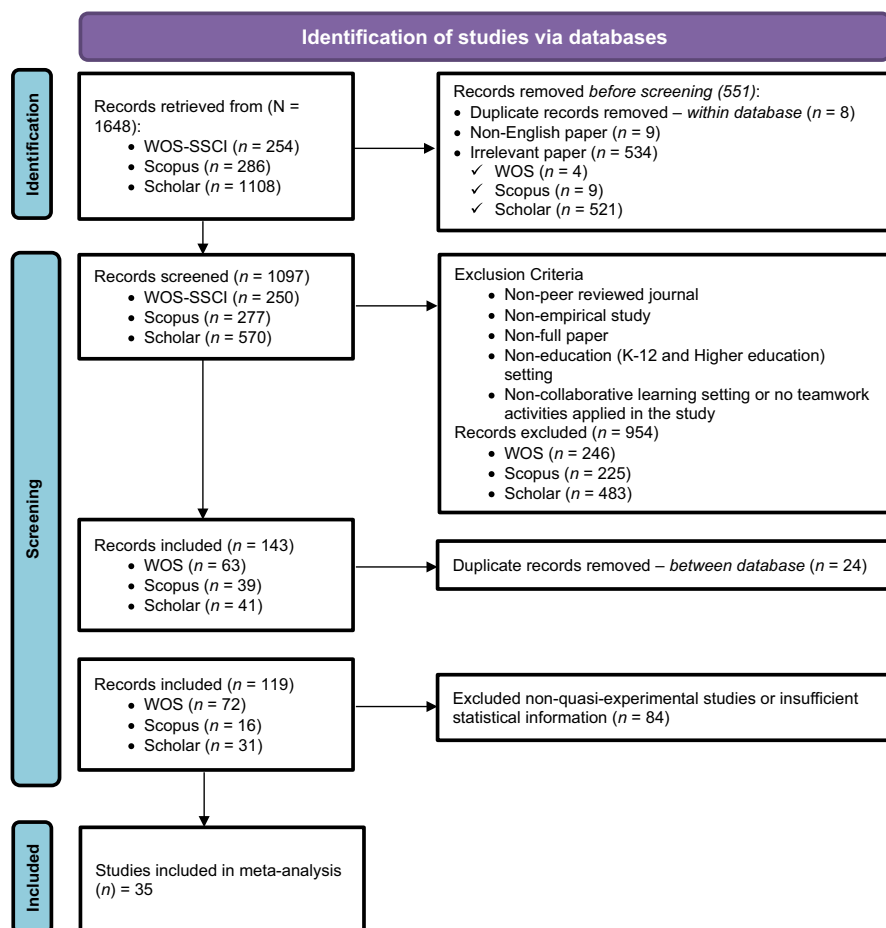


Fig. 1 PRISMA flowchart guiding our process of study selection

(4) non-education settings. Afterward, we conducted a duplication test across the three databases and later removed 24 papers, which resulted in 119 references. For the final step, we include only studies that had sufficient statistical information for meta-analysis.

Learning outcomes

In our study, the included studies must have investigated at least one of three major learning outcomes (i.e., behavior, cognition, and attitudes) (R. Huang et al., 2020; Ritzhaupt et al., 2021; Sailer & Homner, 2020). Behavior learning outcomes constitute learning objectives that measure learners' level of interaction with learning resources, peers, or instructors (Koszalka & Whorway, 2024). These measures can be done through self-assessed questionnaires or direct observations on learners'

overt behaviors, such as log data including time spending and login frequency in the online learning environment. Cognitive learning outcomes are defined as learning objectives that measure learners' comprehension of the knowledge taught (Wei et al., 2021). The measures can be done through objective tests or performance-based tests. Lastly, attitudes or social-emotional of learning cover the spectrum of objectives that measure how learners perceive their learning experiences, whether positive, negative, or neutral emotions affected by learning events (Hachem et al., 2022). The most common measure of this aspect is self-assessed questionnaires. In this meta-analysis, due to a limited number of studies that had investigated behavioral learning outcomes, we opted to eliminate this aspect of learning from our analysis. Thus, this meta-analysis only focuses on two major learning outcomes, including cognitive and attitudinal learning outcomes.

Data analysis

Data extraction

Important categories that were relevant to our study were extracted, including: (1) publisher and year of publication, (2) publisher region, (3) discipline, (4) settings (education level), (5) research methods, (6) research design (between or within subject), (7) sample size, (8) randomization (yes/no), (9) length of experiment, (10) measures, (11) variables (independent and dependent), and (12) statistical information (mean and SD) for control and treatment groups. We divided the workload between two researchers, whose results were then rechecked to ensure consistency and to ensure no crucial information was missed.

Computing effect sizes

Given the sample size of 35 studies, with 28 studies assessing cognitive learning outcomes and 22 studies examining attitudinal learning outcomes, we opted to use Cohen's d for calculating the overall effect size. Cohen's d is widely used when sample sizes are considered sufficient, as it provides a robust measure of effect size across studies. Unlike Hedge's g , which corrects for small sample bias, Cohen's d is appropriate here due to the adequate number of studies included. Cohen (1988, 1992) suggests benchmarks of $d = 0.30$, 0.50 , and 0.80 to represent small, medium, and large effects, respectively. This approach allowed us to reliably interpret the magnitude of effects across the sample. All analyses were conducted using IBM SPSS 29 and R Studio.

Analyses of heterogeneity and publication bias

The heterogeneity in the effect size was measured via two tests, Cochran's Q statistics (Q -test) and I^2 tests. The p -value resulted via Q -test determined whether the presence of heterogeneity in our selected studies. Furthermore, the I^2 tests helped us to determine the proportion of variability in the size caused by heterogeneity

rather than sampling error. We assessed the possibility of publication bias using two methods: visual inspection via a funnel plot and statistical analysis through Egger's and Begg's tests. The funnel plot displayed the distribution of selected studies, with evenly scattered dots around the average effect size indicating no publication bias in study selection. Additionally, Egger's Test provided statistical metrics, including a p -value, to determine the presence of publication bias. A p -value < 0.05 suggested evidence of funnel plot asymmetry and potential publication bias, while a p -value ≥ 0.05 indicated no strong evidence of publication bias in the analysis.

Results

35 publications with sufficient statistical information were included in the analysis. A separate analysis was performed for each learning outcome (cognitive and attitude). The effect size of cognitive learning outcomes was analyzed from 28 studies, and the effect size of attitudinal learning outcomes was analyzed from 22 studies. As we discussed previously, due to the limited number of studies that examined the learning outcomes from a behavior aspect of learning, we omitted the analysis for behavior outcomes. The results were divided into several sections. By extracting relevant information about our selected studies, we first elaborated on their profiles. Secondly, we answered each research question in the following sections.

Description of the included studies

The final review and analysis were carried out on 35 articles. The data extracted from these articles encompassed diverse dimensions, including publisher and publisher region, research design, sample size, field of research or discipline, level of education, data collection instruments, and types of learning outcomes. These selected articles were published from 2015 to 2024. As demonstrated in Table 1, 12 articles (34.3%) were published in 2024, 10 articles (28.6%) were published in 2020, six were published in 2022 (17.1%), and three were published in 2021 (8.6%). One article was published separately in 2015, 2016, 2017, and 2023 (2.9% for each year).

The 35 articles were disseminated across 26 publications. Among these articles, three were published by International Journal of Educational Technology in Higher Education, and three were published by Education and Information Technologies. Computer Assisted Language Learning, Computers & Education, European Physical Education Review, Heliyon, and Learning and Motivation, each of which has published two articles separately. The rest of the articles were distributed in the other 19 peer-reviewed journals, one for each of them. The top three publishing countries of these articles were concentrated on United States ($n=10$), United Kingdom ($n=9$), and Netherlands ($n=8$). Among these studies, over half of them (60%) have been done in higher education (undergraduate and graduate levels) ($n=21$), with the rest 40% being done in k-12 education. In addition, a few studies in our selected corpus reported more than one experiment involving different contexts or a group of participants ($n=14$).

Table 1 Profiles of the 35 selected studies

#	Author(s) & year	Publisher	Region of Publisher	Subject	Education Level	Length of Experiment	Sample Size	Randomization
1	Agustina et al. (2024)	Journal on Efficiency and Responsibility in Education and Science	Czech Republic	Physics and biology	Undergraduate	5 weeks	Experiment 1 CG: 31 EG: 31 Experiment 2 CG: 30 EG: 30	No
2	Alsswey & Malak (2024)	Learning and Motivation	United States	human–computer interaction	Undergraduate	4 months	CG: 87 EG: 89	Yes
3	Bilgin & Gul (2020)	TechTrends	United States	Information technology	Undergraduate	13 weeks	CG: 48 EG: 44	No
4	Chaiyarat (2024)	Thinking Skills and Creativity	Netherlands	Social studies	Junior high school (7th graders)	8 Weeks	CG: 34 EG: 30	No
5	Chang et al., (2024a, 2024b)	Heliyon	Netherlands	nursing students' patient safety education	Undergraduate	September to December 2022	CG: 27 EG: 28	No
6	Chen et al. (2024)	Education and Information Technologies	United States	Mathematics learning	Junior high school (6th graders)	3 weeks	CG: 40 EG: 40	Yes
7	Clarke et al. (2021)	ACM Transactions on Computing Education (TOCE)	United States	Software testing class (Computer Science program)	Undergraduate	A semester length	CG: 60 EG: 62	No
8	de Back et al., (2020a, 2020b)	International Journal of Educational Technology in Higher Education	Netherlands	Neuroanatomy	Undergraduate	40 min each group	CG: 20 EG: 20	Yes

Table 1 (continued)

#	Author(s) & year	Publisher	Region of Publisher	Subject	Education Level	Length of Experiment	Sample Size	Randomization
9	Drissi, Chefrour, Boushaha, & Zarzour (2024)	Education and Information Technologies	United States	Computer science	Undergraduate	3 months	CG: 30 EG: 30	No
10	Dziob (2020a, 2020b)	Res Sci Educ	Netherlands	Physics	High School	2.5 lesson hours (approx. 110 min)	School 1 CG: 31 EG: 37 School 2 CG: 26 EG: 36	No
11	Fan et al. (2015)	Eurasia Journal of Mathematics, Science & Technology Education	Turkey	Biology: Human blood circulation systems	Junior High School (8th grade)	10 Weeks	CG: 23 EG: 23	Yes
12	Fernandez-Rio et al. (2022)	European Physiological Education Review	United Kingdom	Physical education	High School	9 weeks	CG: 27 EG: 27	Yes
13	Ho (2020a, 2020b)	Computer Assisted Language Learning	United Kingdom	English language	Undergraduate	13 weeks	CG: 25 EG: 25	No
14	Huang (2024a, 2024b)	English Teaching & Learning	Taiwan, China	EFL reading	Undergraduate	5 weeks	CG: 52 EG: 52	No
15	Jodoi et al., (2021a, 2021b)	Heliyon	Netherlands	EFL education: Scientific debate	Undergraduate	8 weeks	Experiment 1 CG: 27 EG: 24 Experiment 2 CG: 11 EG: 15	No

Table 1 (continued)

#	Author(s) & year	Publisher	Region of Publisher	Subject	Education Level	Length of Experiment	Sample Size	Randomization
16	Joseph et al. (2024)	Nurse Educator	United States	Healthcare education	Undergraduate	One hour session	CG: 69 EG: 79	No
17	Kim and Kim (2022a, 2022b)	Healthcare	Switzerland	Health and medical science	Undergraduate	8 weeks	CG: 51 EG: 51	Yes
18	Korkmaz and Öztürk (2020)	Participatory Educational Research	Turkey	Social studies	Elementary School (5th grade)	6 weeks	CG: 29 EG: 31	No
19	Lee and Lai (2024)	International Journal of Science and Mathematics Education	Netherlands	Mathematics learning	Elementary School (5th grade)	4 weeks	CG: 26 EG: 26	No
20	Liu (2022a, 2022b)	Computer Assisted Language Learning	United Kingdom	English language	Undergraduate	16 weeks	CG: 30 EG: 30	Yes
21	López-Jiménez et al. (2022)	Computer Methods and Programs in Biomedicine	Ireland	Medical studies: General and descriptive anatomy of the locomotor system (GAHMS) course	Undergraduate	15 weeks	Experiment 1 CG: 70 EG: 80 Experiment 2 CG: 80 EG: 83 Experiment 3 CG: 57 EG: 85 Experiment 4 CG: 83 EG: 86	No

Table 1 (continued)

#	Author(s) & year	Publisher	Region of Publisher	Subject	Education Level	Length of Experiment	Sample Size	Randomization
22	Putz et al., (2020a, 2020b)	Computers in Human Behavior	United Kingdom	Sustainable supply chain management	Secondary and tertiary education	June 2015 to May 2017	Experiment 1 CG: 18 EG: 16 Experiment 2 CG: 73 EG: 261	Yes
23	Raffone (2022a, 2022b)	Language Teaching Research	United Kingdom	English history and culture	secondary school	Only a few hours could be dedicated to the experiment	CG: 12 EG: 83 [22, 19, 10, 22, 10] Note: This study had 1 cohort of control group and 5 cohorts of treatment group	No
24	Rodrigues et al., (2022a, 2022b)*	International Journal of Educational Technology in Higher Education	Netherlands	Computer science	Undergraduate	14 weeks	CG: 138 EG: 659	No
25	Sailer and Sailer (2021a, 2021b)	British Journal of Educational Technology	United Kingdom	Education science and practices	Undergraduate	a semester length	CG: 109 EG: 96	No
26	Sánchez et al. (2020)	Multimodal Technologies and Interaction	Switzerland	Spanish language and literature	High school	NA	CG: 30 EG: 30	Yes
27	Sanina et al. (2020)	Computers & Education	United Kingdom	Social science education: Public Economic Sector	Graduate	NA	CG: 83 EG: 86	No

Table 1 (continued)

#	Author(s) & year	Publisher	Region of Publisher	Subject	Education Level	Length of Experiment	Sample Size	Randomization
28	Sotos-Martínez et al. (2024)	European Physiological Education Review	United Kingdom	Physical education	Primary school	six weeks	CG: 256 EG: 250	No
29	Stansbury and Earnest (2017)	Teaching of Psychology	United States	Industrial/organizational (I/O) psychology	Undergraduate	One semester long	CG: 44 EG: 49	Yes
30	Tan and Hew (2016a, 2016b, 2016c)	Australasian Journal of Educational Technology	Australia	Methods of research and enquiry course: Designing good questionnaire	Graduate	3 days	CG: 11 EG: 11	Yes
31	Treiblmaier and Putz (2020a, 2020b)	Learning and Motivation	United States	Sustainable logistics	High school	October 2015 and June 2016	Exp. 1 CG: 110 EG: 227 Exp. 2 CG: 111 EG: 231	No
32	Tsai et al. (2024)	Journal of Educational Computing Research	United States	Block-based programming	Undergraduate	18 weeks	CG: 40 EG: 43	No
33	Yang et al., (2023a, 2023b, 2023c)	International Journal of Educational Technology in Higher Education	Netherlands	Laboratory safety (fire, electrical, and information)	Undergraduate	NA	CG: 16 EG: 19 EG+GMF: 18***	Yes
34	Zainuddin et al., (2020a, 2020b)	Computers & Education	United Kingdom	Science	High school	3 weeks	CG: 31 EG 1: 33 EG 2: 30	No

Table 1 (continued)

#	Author(s) & year	Publisher	Region of Publisher	Subject	Education Level	Length of Experiment	Sample Size	Randomization
35	Zhang and Crawford (2024)	Education and Information Technologies	United States	EFL course: Scientific debate	Elementary school	20 Weeks	CG: 22 EG: 23	No

*The study only examined behavior outcomes as the dependent variable. Thus, it's excluded in the analysis
**CG control group, EG experiment group
***This study examined the effects of multimedia learning. Three groups were involved, CG control group, EG experiment group (Multimedia Learning), and EG + GMF experiment group (multimedia learning with gamification features)

These articles were distributed across eight subject areas including education ($n=13$, 37.1%), computer science ($n=6$, 17.1%), medical science ($n=5$, 14.3%), social sciences ($n=4$, 11.4%), natural sciences ($n=3$, 8.6%), business ($n=2$, 5.7%), laboratory safety ($n=1$, 2.9%), and research methodology ($n=1$, 2.9%). It can be observed that most of the research concerning gamification in collaborative learning has been conducted in education, computer science, and medical science.

Among the 35 studies on gamification in collaborative learning, up to 26 of them (74.3%) applied a quasi-experimental approach with a control group to detect the gamification effects in collaborative learning while the rest nine (25.7%) conducted a mixed-method approach, which indicates a strong preference for quantitative methodology for the study of this topic. No study solely employs qualitative methods. Concerning the research design of these studies, 34 of them conducted within-subject method, or independent measures design, where experimental and control groups of participants are exposed to different levels or conditions of the independent variable. Only one study adopted a between-subjects design. Therefore, it can be concluded that studies on gamification in collaborative learning within the past 10 years have an obvious tendency towards quantitative methodology, among which, quasi-experimental design has become the most frequently used method in conducting quantitative research on this topic. Concerning quasi-experimental design, the sample size varies from study to study, ranging from as small as 11 participants in a single group to as large as over 200 participants.

The data collection instruments in these studies include tests (including pre and post-tests, exams, quizzes, and other types of assessment), questionnaires, interviews, psychometric scales, surveys, and log data. Collecting data through tests was the most widely used instrument and it has been adopted by 32 out of 35 studies. Following that were the instruments of questionnaires and modified psychometric scales, which have been employed by 13 and eight studies, separately. Seven studies conducted interviews and six employed surveys to collect data; Only two research mentioned using log data stored in the learning platforms and one using self-reflection journals as their data resource.

RQ#1: What types of learning outcomes did researchers investigate when using gamification in collaborative learning?

We analyzed 35 selected studies within our corpus to explore the focal points of researchers' inquiries into the roles of gamification in collaborative learning. In summary, three primary academic outcomes were scrutinized: cognitive outcomes, behavioral outcomes, and attitude or social-emotional outcomes (Fig. 2). While some studies offered comprehensive examinations of all three outcomes within a single study, others adopted a more focused approach, delving into specific types of outcomes (Fig. 3). A diverse array of measurement methods was employed. Cognitive outcomes were predominantly assessed through knowledge tests (Clarke et al., 2022a; De Back et al., 2020a, 2020b; Dziob, 2020a, 2020b; Putz et al., 2020a), although a few studies employed alternative approaches such

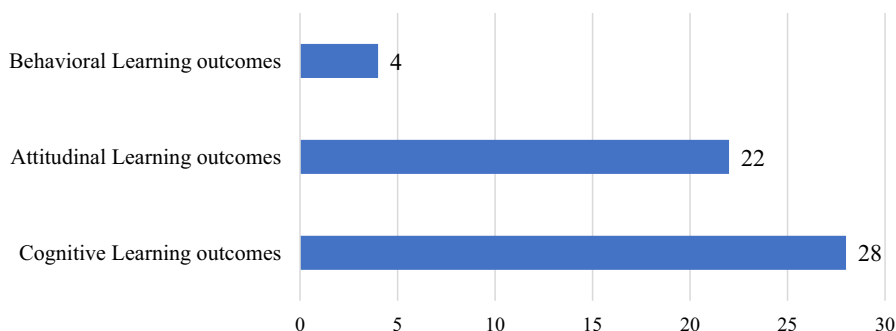


Fig. 2 The frequencies of learning outcomes being studied in our selected corpus

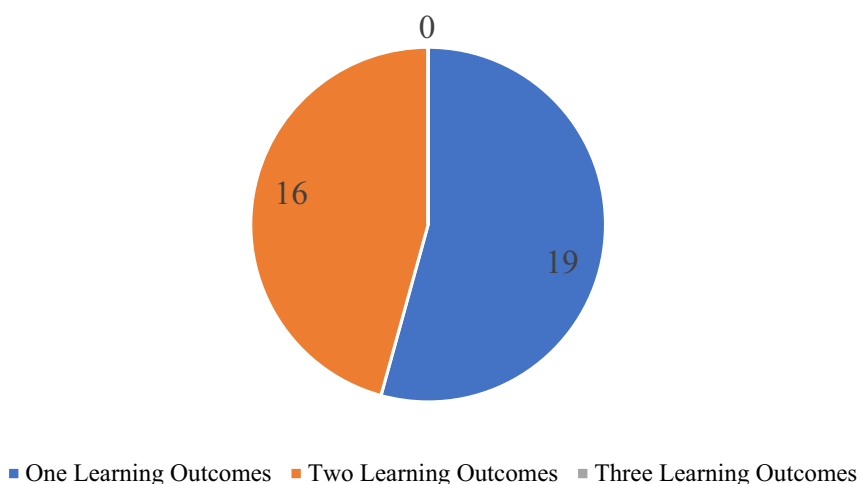


Fig. 3 The number of learning outcomes being studied in each study

as performance-based tests (e.g., writing performance) (J. Ho, 2020a, 2020b) and artifact-based measurements (M. Tan & Hew, 2016a).

Regarding attitude outcomes, despite encompassing various measurement aspects (e.g., cognitive load perception, learning motivation, and perceived use of gamification), researchers heavily relied on self-assessed questionnaires (Pozo Sánchez et al., 2020a, 2020b; Treiblmaier & Putz, 2020a, 2020b; Q.-F. Yang et al., 2023a). Lastly, behavioral aspects were the least investigated variables within our selected corpus (Jodoi et al., 2021a, 2021b; Rodrigues et al., 2022a, 2022b; M. Tan & Hew, 2016a). Studies exploring the behavioral aspects of learning typically depended on log data from systems, including metrics such as length of access, IDE usage, attempts, login data, and number of posts. Unfortunately, one study did not have sufficient statistical information to be included in the analysis. Thus, we eliminated behavioral outcomes analysis in our meta-analysis due to limited number of cases (two studies).

Overall effect size on cognitive learning outcomes

Forest Plot

■ Effect size of each study
◆ Estimated overall effect size
▬ Estimated overall confidence interval

Study	Effect Size (approx.)	Lower CI (approx.)	Upper CI (approx.)
de Haan et al. (2024)	0.36	0.21	0.49
Alotaibi & Malik (2024)	1.15	0.71	1.59
Wojcik & Dui (2020)	0.51	0.01	1.01
Chaignant (2024)	0.50	0.34	0.66
Cheng et al. (2024)	0.72	0.26	1.17
Chen et al. (2024)	0.44	0.08	0.80
Clarke et al. (2023)	0.73	0.09	1.37
de Haan et al. (2020)	1.06	0.52	1.40
Dufresne et al. (2024)	0.45	0.29	1.00
Deitch (2020)	1.13	0.51	1.75
Pan et al. (2015)	1.30	0.49	1.91
Hu (2020)	0.47	0.08	0.86
Maang (2024)	-0.15	-0.70	0.20
Joshi et al. (2023)	0.10	-0.36	0.41
Wu & Kim (2022)	0.44	0.32	1.37
Kurkova & Gotsch (2020)	0.68	0.10	1.21
Lee & Lai (2024)	0.87	0.44	1.51
Liu (2022)	0.85	0.49	1.01
Edgar-Jimenez et al. (2024)	1.30	1.05	1.51
Puts et al. (2020)	1.34	0.70	1.97
Haffner (2022)	2.77	2.30	3.15
Salinas & Salinas (2023)	1.53	1.16	1.85
Sanjiva et al. (2020)	-0.87	-1.10	-0.45
Stannbury & Barnett (2022)	0.63	-0.04	1.70
Yeo & New (2014)	0.16	0.01	1.20
Yan et al. (2024)	0.00	-0.47	0.67
Raimondini et al. (2020)	0.01	-0.28	0.29
Xiang & Crawford (2024)	1.11	0.48	1.74
Overall	0.08	-0.52	1.25

Model: Random-effects model
 Heterogeneity: Tau-squared = 0.88, I-squared = 21.92, Q-squared = 0.95
 Test of overall effect size: $Z = 4.80$, p -value = 0.00

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Publication bias assessments on cognitive learning outcomes

A funnel plot was generated to investigate potential publication bias in our analysis on the influence of gamification on cognitive learning outcomes. According to the visual inspection, we found that the figure showed symmetrical distribution indicating no evidence of publication bias on our selected corpus. For further investigation, we confirmed the findings with statistical analysis using Egger's and Begg's tests. The publication bias assessments, conducted through Egger's and Begg's tests, provide valuable insight into potential biases in our meta-analysis investigating the impact of gamification on learners' learning outcomes in collaborative learning contexts. Egger's test, employing a mixed-effects meta-regression model with the standard error as the predictor, revealed no statistically significant evidence of funnel plot asymmetry ($z=0.5434371$, $p=0.5914583$) (Fig. 5).

Furthermore, Begg's test was conducted to assess the presence of publication bias in meta-analysis. The p -value obtained from the test was 0.174, indicating that there is no significant evidence of publication bias ($p \geq 0.05$). This suggests that the effect sizes in the included studies do not exhibit systematic bias, and the results of the meta-analysis are likely to be robust. While these results provide reassurance regarding the robustness of our findings regarding publication bias, we acknowledge the importance of continued vigilance in assessing other potential sources of bias and interpreting the results within the broader context of the study.

Overall effect size on attitudinal learning outcomes

Gamification has emerged as a prominent strategy for enhancing learners' attitudes outcomes within collaborative learning environments. In this random-effects meta-analysis involving 22 studies, the overall effect size was found to be statistically

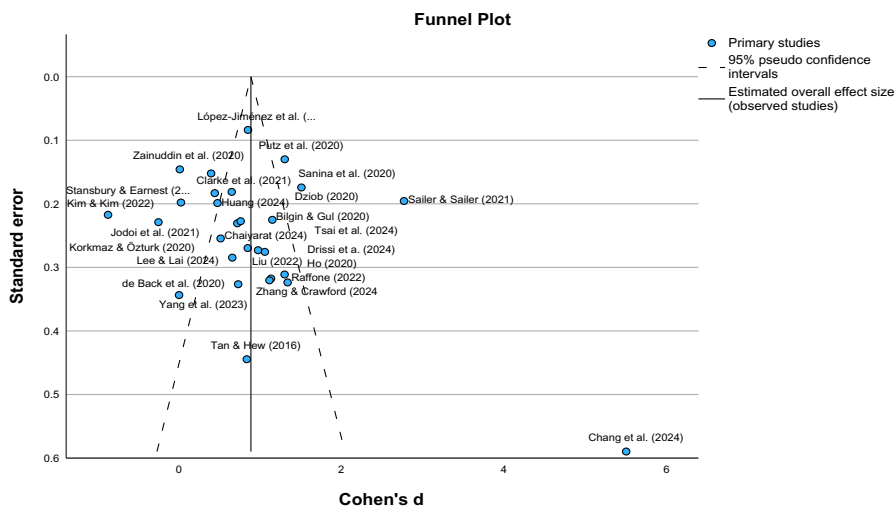


Fig. 5 Funnel plot of the effects of gamification on cognitive learning outcomes in collaborative learning

significant, with an estimated Cohen's d of 0.748 (SE=0.159, $p < 0.0001$). This indicates a positive impact of the intervention on cognitive learning outcomes, with a confidence interval ranging from 0.436 to 1.060, suggesting that the true effect is both meaningful and substantial. The analysis revealed a high level of heterogeneity among the studies, characterized by an estimated τ^2 of 0.498 (SE=0.172) and an I^2 statistic of 93.10%, indicating that a significant portion of the variability in effect sizes can be attributed to genuine differences between studies rather than random sampling error. The heterogeneity test ($Q(df=21)=299.313$, $p < 0.0001$) confirmed the presence of significant variability, underscoring the need for further exploration into potential moderators that could influence the observed effects across different contexts and populations (Fig. 6).

Publication bias assessments on attitudinal learning outcomes

Funnel plot generated from the analysis showed symmetrical distributions around the weighted mean effect size. According to this visual inspection, symmetrical funnel plot proves that there was no publication bias. However, the visual inspection alone was not sufficient to examine the presence of publication bias. The evaluation of publication bias in the meta-analysis revealed significant findings from both Egger's test and Begg's test. Egger's test yielded a z -value of 3.13 ($p = 0.0053$), indicating significant evidence of publication bias ($p < 0.05$). This suggests that smaller studies with positive results may be disproportionately represented in the meta-analysis, potentially skewing the overall effect size. The estimate of the intercept when the standard error approaches zero was found to be -0.5185, with a 95% confidence interval ranging from -1.1193 to 0.0822. Similarly, Begg's test provided a p -value of 0.0118, further supporting the presence of significant publication bias ($p < 0.05$).

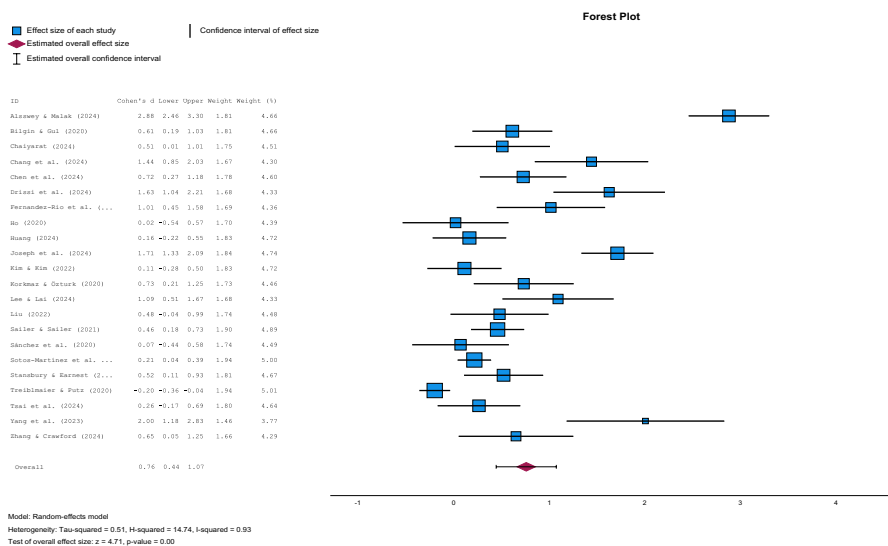


Fig. 6 Forest plot of the effects of gamification on attitude learning outcomes in collaborative learning

These results underscore the importance of considering potential biases in the interpretation of the findings, highlighting the necessity for additional investigations to identify the sources of bias and to implement strategies that mitigate their influence on the study's conclusions (Fig. 7).

These findings collectively underscore the importance of considering the impact of publication bias on the validity and reliability of our meta-analysis results. The significant results from both tests suggest that caution should be exercised when interpreting the attitudinal learning outcomes of our study in particular.

RQ#3: How did the length of experiment and education level influence the outcomes of the studies?

We found that Q -tests and I^2 output for the effects of gamification on cognitive and attitudinal learning outcomes showed a substantial and significant level of heterogeneity and it is recommended that a subgroup analysis was conducted to delve into the factors that might influence this heterogeneity. We used the length of experiment and education level to run the subgroup analysis. It was posited that the duration of an experiment could significantly impact its outcomes, with longer interventions being more likely to yield positive results (Cusser et al., 2021). Previous literature has indicated that gamification research often suffers from insufficient intervention time, thereby limiting the realization of potential benefits or the efficacy of the approach (Rajanen & Rajanen, 2019; Sanchez et al., 2020). In addition, research has found that the influences of gamification vary across different levels of education. Thus far, gamification research has predominantly focused on higher education levels (Andrade et al., 2020b; J. Kim & Castelli, 2021), despite significant findings emerging from studies conducted in elementary and secondary education. Another

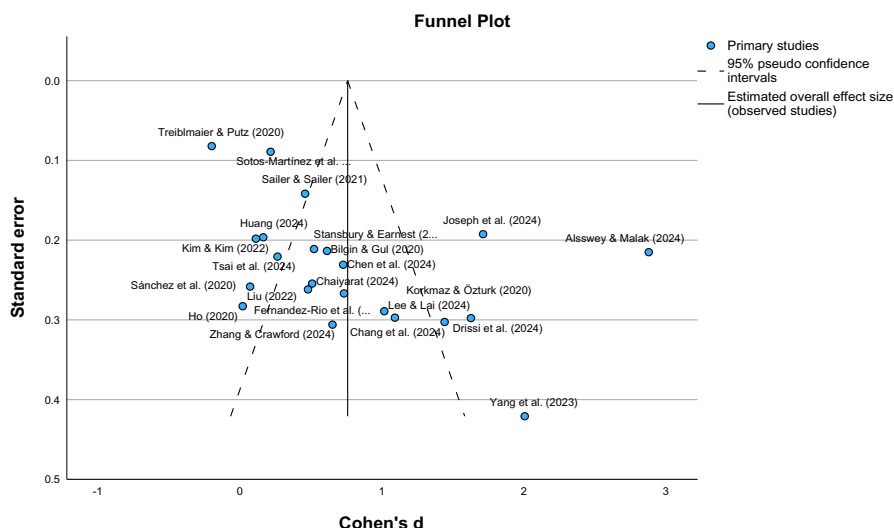


Fig. 7 Funnel plot of the effects of gamification on attitude learning outcomes in collaborative learning

subgroup analysis examined whether education level significantly influenced the effects of gamification on learning outcomes in collaborative learning.

Length of experiment

Sub-group analysis in cognitive learning outcomes

The analysis of the effects of gamification on cognitive learning outcomes was conducted based on the duration of the experiments, which were categorized into short (less than 5 weeks), medium (5 to 8 weeks), and long lengths (more than 8 weeks). The results demonstrate a notable variation in the effect sizes across these different durations. For the short-length experiments, the effect of gamification on cognitive learning outcomes was significant, with an effect size of 0.695 ($SE=0.1253$, $Z=5.543$, $p<0.001$). The 95% confidence interval (CI) for this effect ranged from 0.449 to 0.940, indicating a moderate to large positive effect. This suggests that short-term exposure to gamified learning environments can lead to meaningful improvements in cognitive outcomes. In contrast, the medium-length experiments yielded a smaller effect size of 0.341 ($SE=0.1754$, $Z=1.945$, $p=0.052$), which approached but did not reach statistical significance. The 95% CI ranged from -0.003 to 0.685, indicating that the effect of gamification in medium-length experiments was less consistent and possibly negligible, as the confidence interval crossed zero. This implies that the cognitive benefits of gamification may not be as robust in experiments of intermediate duration.

The most substantial effect was observed in long-length experiments, where the effect size was 1.205 ($SE=0.3046$, $Z=3.957$, $p<0.001$). The 95% CI ranged from 0.608 to 1.802, reflecting a large and statistically significant effect. This finding suggests that extended engagement with gamified learning environments significantly enhances cognitive learning outcomes, potentially due to sustained interaction with gamification elements that may reinforce learning over time. Overall, across all studies, the combined effect of gamification on cognitive learning outcomes was significant, with an overall effect size of 0.884 ($SE=0.1843$, $Z=4.796$, $p<0.001$). The 95% CI ranged from 0.523 to 1.245, indicating a large positive effect. This suggests that, on average, gamification exerts a substantial impact on cognitive outcomes, particularly when experiments are conducted over longer periods.

Sub-group analysis in attitudinal learning outcomes

A subgroup analysis was conducted to investigate the effects of gamification on attitudinal learning outcomes, categorizing the experimental durations into short, medium, and long periods. The results demonstrate varying effects based on the length of exposure to gamification. For the short-period experiments, the effect size was 0.847 ($SE=0.4788$, $Z=1.769$, $p=0.077$), with a 95% confidence interval (CI) ranging from -0.091 to 1.785. Although this effect size suggests a moderate positive impact of gamification on attitudinal outcomes, the result was not statistically significant, as indicated by the p -value and the confidence interval, which includes

zero. This suggests that for short-term interventions, the effect of gamification on learners' attitudes may be uncertain or less robust.

The medium-period experiments demonstrated a statistically significant effect, with an effect size of 0.542 ($SE=0.2426$, $Z=2.233$, $p=0.026$), and a 95% CI of 0.066 to 1.017. This finding indicates that gamification has a positive and measurable effect on attitudinal learning outcomes when the intervention is of moderate duration. The fact that the confidence interval does not include zero further supports the reliability of this effect. In the long-period experiments, a larger and statistically significant effect size of 0.824 ($SE=0.2275$, $Z=3.620$, $p<0.001$) was observed, with a 95% CI ranging from 0.378 to 1.270. This result suggests that long-term exposure to gamification produces a strong positive impact on attitudinal learning outcomes. The narrow confidence interval indicates a high degree of precision in this estimate, underscoring the substantial benefits of prolonged engagement with gamification in shaping learners' attitudes.

The overall analysis across all studies yielded an effect size of 0.757 ($SE=0.1606$, $Z=4.713$, $p<0.001$), with a 95% CI of 0.442 to 1.071. This significant overall effect suggests that gamification has a meaningful and positive influence on attitudinal outcomes across varying time periods. However, the effect is more pronounced in medium and long-duration interventions compared to short-term ones.

Level of education

Sub-group analysis in cognitive learning outcomes

A subgroup analysis was conducted to evaluate the impact of gamification on cognitive learning outcomes across different levels of education: elementary, secondary, and tertiary (or higher) education. The results reveal significant positive effects of gamification on cognitive outcomes across all educational levels, with some variation in the magnitude of the effect sizes. For learners in elementary education, the effect size was 0.806 ($SE=0.1353$, $Z=5.954$, $p<0.001$), with a 95% confidence interval (CI) ranging from 0.540 to 1.071. This finding indicates a large and statistically significant effect, suggesting that gamified learning environments have a robust positive impact on cognitive learning outcomes for elementary-level students.

At the level of secondary education, the effect size was 0.798 ($SE=0.2236$, $Z=3.567$, $p<0.001$), with a 95% CI of 0.359 to 1.236. This effect, also statistically significant, indicates a similarly strong impact of gamification on cognitive outcomes for secondary-level students. The slightly wider confidence interval suggests a moderate degree of variability, but the effect remains substantial. For learners in tertiary or higher education, the effect size was 0.940 ($SE=0.2951$, $Z=3.184$, $p=0.001$), with a 95% CI between 0.361 and 1.518. This result reflects the largest effect among the educational subgroups, suggesting that gamification has a particularly strong impact on cognitive outcomes for higher education students. However, the wider confidence interval indicates more variability in the effect at this level of education. The overall analysis across all education levels showed an effect size of 0.884 ($SE=0.1843$, $Z=4.796$, $p<0.001$), with a 95% CI of 0.523 to 1.245. This

significant overall effect highlights the considerable positive influence of gamification on cognitive learning outcomes, regardless of the educational level. However, the results suggest that gamification may be especially effective in higher education settings.

Sub-group analysis in attitudinal learning outcomes

A subgroup analysis was conducted to examine the effects of gamification on attitudinal learning outcomes across different educational levels: elementary, secondary, and tertiary (or higher) education. The results indicate significant variations in the effect sizes, highlighting the differential impact of gamification based on the learners' educational background. For elementary education, the effect size was 0.618 ($SE=0.1660$, $Z=3.724$, $p<0.001$), with a 95% confidence interval (CI) ranging from 0.293 to 0.943. This finding suggests a moderate yet statistically significant positive effect of gamification on the attitudes of elementary learners, indicating that gamified learning environments can effectively enhance their engagement and motivation.

In contrast, for secondary education, the effect size was 0.307 ($SE=0.2682$, $Z=1.144$, $p=0.252$), with a 95% CI of -0.219 to 0.832 . This result did not reach statistical significance, suggesting that the impact of gamification on attitudinal outcomes for secondary-level students may be less reliable or substantial. The confidence interval that includes zero further indicates a lack of conclusive evidence for the effectiveness of gamification at this educational stage. For learners in tertiary or higher education, the effect size was notably higher at 0.930 ($SE=0.2468$, $Z=3.767$, $p<0.001$), with a 95% CI between 0.446 and 1.413. This result demonstrates a strong and statistically significant positive effect of gamification on attitudinal learning outcomes, indicating that higher education students are particularly responsive to gamified learning strategies, which may enhance their attitudes toward the learning process.

The overall analysis yielded an effect size of 0.757 ($SE=0.1606$, $Z=4.713$, $p<0.001$), with a 95% CI of 0.442 to 1.071. This significant overall effect suggests that gamification positively influences learners' attitudes across various educational levels, although the strength of this effect varies, with notable effectiveness in elementary and tertiary education settings.

Discussions

This meta-analysis examines the effects of gamification on learning outcomes in collaborative learning settings. According to our study, we were able to confirm the positive effect size of gamification on learning outcomes when applied in collaborative learning environments. By comparing our data with existing studies, we observed consistent positive trends in cognitive and attitudinal learning outcomes (J. C.-S. Ho et al., 2022; R. Huang et al., 2020; J. Kim & Castelli, 2021). This alignment with prior research underscores the robustness and reliability of gamification as an educational tool. To fill the gaps evident in previous meta-analyses on

gamification research, our meta-analysis focused on the utilization of gamification in collaborative learning. Our goals were to see what types of learning outcomes that had been investigated by scholars when using gamification in collaborative learning and to what extent gamification can improve learning outcomes. Using this meta-analysis, we can better understand what aspects of learning were commonly investigated in collaborative learning gamification research. Furthermore, we measured the effect sizes of gamification on cognitive and attitudinal learning outcomes to inform future research about the meaningfulness of gamification's impact in collaborative learning environments. By providing these measures, we aim to help compare the effectiveness of different interventions, inform policy and practice, and identify broader trends in research. In addition, we delve deeper into how the efficacy of gamification can be varied across length of implementation and level of education.

For cognitive learning outcomes, we found positive impacts of gamification on cognitive learning outcomes with the effect size (d) of 0.875, $p < 0.0001$ and high probability of substantial heterogeneity ($Q(df=27)=350.711$, $p < 0.0001$). Our meta-analysis confirms previous research that reported positive impacts of gamification on academic achievement in collaborative learning environments (Clarke et al., 2022b; da Rocha Seixas et al., 2016; Uz Bilgin & Gul, 2020a). By synthesizing data from multiple studies, we observed that gamification consistently enhances learners' academic achievement when it is combined with proper instructional designs, such as collaborative or cooperative learning (de-Marcos et al., 2016; J. W.-T. Huang, 2024a, 2024b; Uz Bilgin & Gul, 2020b). These studies found that the presence of gamification has become the catalyst to improve collaboration within group contexts. Incorporating gamification in collaborative learning also leads to other benefits, such as reducing cognitive load (Q.-F. Yang et al., 2023b), increasing interactions between team members (Liu, 2022a, 2022b; Raffone, 2022a, 2022b), and motivating learners to learn more complex learning activities (M. Tan & Hew, 2016b), which all of these factors lead to better cognitive learning outcomes (academic achievement).

To understand the effect of gamification, one must recognize that, similar to other instructional interventions, there are complex mechanisms that explain its impact. The influence of gamification on cognitive outcomes is often mediated by other factors that are theoretically relevant to these outcomes (De Back et al., 2020a, 2020b; Landers, 2014). This complexity underscores the necessity to consider the interplay of various elements, such as student motivation, engagement, and the specific design of gamified activities, which together contribute to the observed cognitive benefits. It is also imperative to understand that despite the variety of cognitive assessment utilized in out selected studies, knowledge tests (memorization and recalling) are the most popular assessment by researchers (Clarke et al., 2022b; Dziob, 2020a, 2020b; Liu, 2022a, 2022b; Putz et al., 2020b). Some other approaches are reading assessment (J. W.-T. Huang, 2024a, 2024b), writing performance (J. Ho, 2020a, 2020b), and problem-solving (H. Kim & Kim, 2022a, 2022b; Sailer & Sailer, 2021a, 2021b). Thus, it still remains relevant for further research to utilize more complex and authentic assessment to examine the extent to which the efficacy of gamification can be benefiting more complex learning outcomes.

In terms of attitudinal learning outcomes, the results also showed positive impacts of gamification on affective learning outcomes ($d=0.748$, $p<0.0001$) and we also found substantial heterogeneity among the selected studies ($Q(df=21)=299.313$, $p<0.0001$). Despite the positive outcomes on attitudinal outcomes, research has been varied regarding the specific aspect of attitudinal outcomes that they have targeted. Some of those are attitude towards courses (Bilgin & Gul, 2020; Korkmaz & Ozturk, 2020; Stansbury & Earnest, 2017), perceived experience and learning (Stansbury & Earnest, 2017), attitude towards gamification (Dziob, 2020a, 2020b; Jodoi et al., 2021a, 2021b), attitude towards group environments, team member evaluations, group cohesion, (Bilgin & Gul, 2020), and intrinsic motivation and psychological need satisfaction (Sailer & Sailer, 2021a, 2021b; Treiblmaier & Putz, 2020a, 2020b).

One significant factor contributing to the positive perception of gamification is its novelty, offering a unique and engaging approach distinct from traditional learning methods used in control groups within our selected studies. Learners experienced increased motivation and determination to engage in gamified experiences, as they found the course content not only exciting but also highly relevant to their interests (An, 2023). Incorporating gamification in collaborative learning also provides clear goals for teams to strive for (Chang et al., 2024a, 2024b; Landers et al., 2017). By having clear goals in collaborative learning, working in team is more effective and each team member has clear commitment and ways to contribute to their team progress (Landers et al., 2017).

These findings suggest that while gamification generally leads to improved cognitive and affective learning outcomes, the extent of this effect may depend on various factors such as the level of education and the design of implementation. Further research exploring these moderators could provide valuable insights for the effective implementation of gamification strategies in educational settings. We used length of experiments and education levels as the criteria for investigation. Based on the effect sizes output, studies with long-length experiments demonstrated the highest effect size in the aspect of cognitive learning outcomes with $d=1.205$ ($p<0.001$) and $d=0.824$ ($p<0.001$) for attitudinal learning outcomes, indicating pronounced efficacy over extended durations. Following this, studies conducted in comparatively medium length of experiment tended to exhibit a weaker effect size (cognitive learning outcomes: d (0.341), $p=0.052$; attitudinal learning outcomes: d (0.542), $p=0.026$) compared to short and long length experiment. It is also found that the outcomes in the medium length of experiment are found to be not statistically significant, implying limited effectiveness within moderate durations.

Our study supports partially the findings of a recent meta-analysis, which concluded that gamification interventions lasting three months or less are more effective in enhancing learning outcomes (Yu et al., 2024). This finding confirmed a previous study (Rodrigues et al., 2022a, 2022b) that found gamification suffered from a novelty effect but also can benefit from familiarization effects that followed a U-shaped pattern over time. Thus, longitudinal or long-length interventions were needed to observe the dynamics of gamification effects (Ennis et al., 2016; Stepanovic & Mettler, 2018).

In addition, the effectiveness of gamification on learning outcomes varied across different educational levels, with notable differences in both cognitive and attitudinal outcomes. The subgroup analysis on cognitive learning outcomes across elementary, secondary, and tertiary education levels demonstrated significant positive effects of gamification at all levels. In the aspect of cognitive learning outcomes, studies in higher education gained the highest effect size compared to elementary and secondary education level, which was $d=0.940$ ($SE=0.2951$, $Z=3.184$, $p=0.001$), with a 95% CI ranging from 0.361 to 1.518. This result reflects the largest effect size among the educational subgroups, indicating that gamification had the strongest impact on cognitive outcomes at this level, though the wider confidence interval suggests more variability in outcomes.

Regarding attitudinal learning outcomes, similar to cognitive learning outcomes, the studies conducted in higher education gained the highest effect compared to studies performed in elementary and secondary education levels ($d=0.930$; $SE=0.2468$, $Z=3.767$, $p<0.001$), with a 95% CI of 0.446 to 1.413, reflecting a strong and significant positive impact of gamification on learners' attitudes. This finding highlights that higher education students, in particular, respond well to gamified learning strategies, which appear to enhance their attitudes toward the learning process. These findings align with previous research suggesting that while gamification is particularly effective in higher education (J. Kim & Castelli, 2021), there is a need for more studies on its impact at the elementary and secondary levels, where its influence on learning outcomes remains underexplored (Behl et al., 2022; Oliveira et al., 2022).

Study limitations and recommendations

Understanding how gamification influences behavioral engagement in collaborative learning settings is of paramount importance for educational research and practice. Regrettably, our review revealed a notable oversight in literature, with many studies failings to adequately investigate behavioral engagement within collaborative learning environments. In our study, it became evident that scholars have not yet prioritized the investigation of behavioral outcomes in collaborative learning. Furthermore, we observed a scarcity of studies that specifically explore the interaction patterns or behavioral change in collaborative learning influenced by gamification. Potential measurements, such as protocol analysis techniques to quantify behavioral changes (Kan & Gero, 2011) and system-based measures (Collazos et al., 2007), can be leveraged to solidify our understanding of the roles of gamification on learners' behavioral changes in collaborative learning. Recognizing the pivotal role of learner interaction in these contexts is crucial for fostering positive learning experiences and achieving desired cognitive outcomes. It is imperative for researchers and practitioners alike to prioritize the examination and enhancement of behavioral data within log systems to facilitate accurate predictions of learner engagement in gamified collaborative learning environments.

In the context of meta-analysis studies, methodological rigor is a valid concern for readers and researchers alike. In our investigation, it is essential to acknowledge

the limitation stemming from the absence of a systematic quality assessment of the studies included. While systematic quality assessments, such as the application of tools like the checklist for the assessment of methodological quality (Downs & Black, 1998), are commonly utilized to evaluate the methodological soundness and risk of bias in individual studies, our analysis did not comprehensively integrate such assessments. This limitation arose due to the relatively limited number of relevant studies ($n=35$) available to address our research inquiries regarding the impact of gamification on learning outcomes in collaborative environment. Consequently, we were unable to quantitatively evaluate the quality of each study included or gauge the influence of potential biases on our overall findings. Notwithstanding this limitation, we contend that our analysis offers valuable insights into the role of gamification in collaborative learning contexts. However, caution is warranted when interpreting the results, and future research endeavors should strive to address this limitation through more comprehensive and systematic quality assessments.

Conclusions

Our findings asserted the positive effects from incorporating gamification for improving learning outcomes in collaborative learning. In our corpus, gamification was shown to potentially improve cognitive and attitudinal aspects of learning as compared to collaborative learning alone. However, we recommend that the results must be considered with some careful consideration, such as the length of implementation and education level. In addition, there were some limitations regarding methodological rigors that need to be fully understood.

Along with previous studies and meta-analyses, we found positive impacts from implementing gamification in collaborative learning for cognitive outcomes measured by knowledge-, performance-, or artefact-based tests. In terms of attitudinal learning outcomes, the results also showed positive outcomes from incorporating gamification in collaborative learning. In the scope of attitude outcomes, there were multifaceted aspects of attitudinal outcomes that researchers have examined, such as learning motivation, mind-flow experience, perceived use of gamification, and perceived engagement. According to our findings, there were only a few studies that examined how gamification influenced the quality of interaction among learners in collaborative activities. Several aspects recorded in the corpus were perceived cooperation, perceived interaction, peer collaboration perception, and attitude towards group cohesion. However, for the studies that examined the attitude outcomes, we found the possibility of publication bias in the corpus indicating the selected studies did not represent the true distribution of the studies.

It is important to acknowledge the variability in effect sizes observed across the selected studies. To gain a deeper understanding of these variations, we conducted subgroup analyses. These analyses offered additional insights into how factors such as the duration of the experimental interventions (short, medium, and long) and the educational levels of participants (elementary, secondary, and tertiary or higher education) may influence the efficacy of gamification. Our findings showed that the length of experiments influenced the effect size, indicating the longest intervention

resulted in the greatest effect sizes compared to short or medium length of experiment. On the other aspect, despite the significant heterogeneity in the effect of gamification, we found moderate to large effect size across different levels of education, including elementary, secondary, and higher education levels. Given the complexity of these relationships, further comprehensive investigations are imperative to fully elucidate the impact of gamification on different conditions or education levels.

The present study endeavors to shed light on the current application of gamification in collaborative learning contexts and its influence on learning outcomes. While extant research suggests positive outcomes associated with gamification, our findings underscore the pressing need for further exploration of additional facets of learning outcomes crucial for fostering successful collaborative learning experiences. An important avenue for future research involves investigating the impact of gamification on dimensions such as teamwork quality, dynamics, and group cohesion, viewed through the lens of behavioral outcomes. Furthermore, there is a call for more holistic investigations that examine the interplay between gamification and all dimensions of learning outcomes, including cognitive, behavioral, and attitudinal aspects. Such research will hold promises for elucidating the multifaceted effects of gamification in collaborative learning settings.

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Data availability The datasets generated and/or analyzed during the current study are not publicly available, but they are available from the corresponding author at reasonable request.

Declarations

Competing interests The authors declare no conflict of interest.

Ethical Statement We affirm that this manuscript is an original work by us and has not been published or submitted elsewhere. The study adheres to ethical standards in scientific research and publication, ensuring transparency and replicability. Additionally, the manuscript has undergone a rigorous peer review process in compliance with ethical publishing guidelines.

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